



Chapter 4

Applications of Derivatives

MAE101 - CALCULUS

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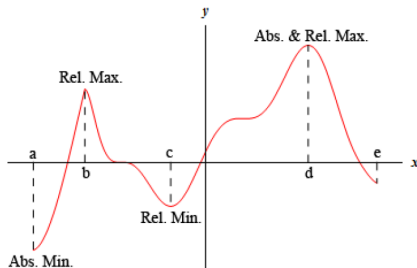
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Max & Min

1 Max & Min

Definition

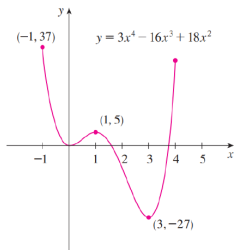
1. The function f has an **absolute (global) minimum** at c if $f(c) \leq f(x)$, $\forall x \in D$.
2. The function f has an **absolute (global) maximum** at c if $f(c) \geq f(x)$, $\forall x \in D$.
3. The function f has an **relative (local) maximum** at c if $f(c) \geq f(x)$, when x near c
4. The function f has a **relative (local) minimum** at c if $f(c) \leq f(x)$, when x near c .
where D is the domain of f .



Example

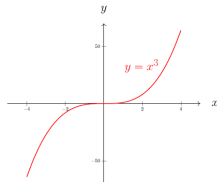
1 Max & Min

Example 1.



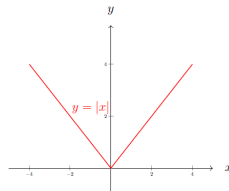
$$f(x) = 3x^4 - 16x^3 + 18x^2, \\ x \in [-1, 4]$$

Example 2.



$f'(0) = 0$ but f has no max and min (global and local) at $x = 0$.

Example 3.



If $f(x) = |x|$ then $f(0) = 0$ is a min value but $f'(0)$ does not exist.



Critical number

1 Max & Min

Definition

A **critical number** (điểm tới hạn) of a function is a number c in the domain of f such that **either** $f'(c) = 0$ or $f'(c)$ does not exist.

Example. Find the critical numbers of $f(x) = x^{3/5}(4 - x)$.

Critical number

1 Max & Min

Definition

A **critical number** (điểm tới hạn) of a function is a number c in the domain of f such that **either** $f'(c) = 0$ or $f'(c)$ does not exist.

Example. Find the critical numbers of $f(x) = x^{3/5}(4 - x)$.

Solution. The Product Rule gives

$$f'(x) = x^{3/5}(-1) + (4 - x) \left(\frac{3}{5} x^{-2/5} \right) = \frac{12 - 8x}{5x^{2/5}}.$$

Therefore,

- $f'(x) = 0$ if $12 - 8x = 0$, that is, $x = \frac{3}{2}$
- $f'(x)$ does not exist when $x = 0$.

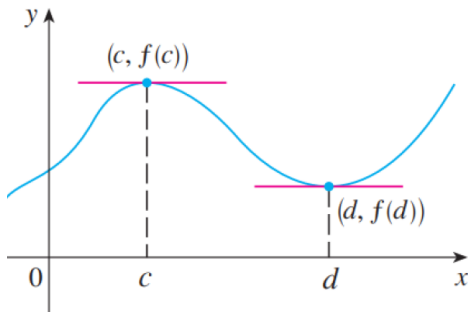
Thus, the critical numbers are $\frac{3}{2}$ and 0 .

Fermat's theorem

1 Max & Min

If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

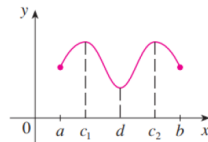
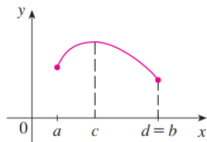
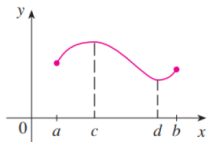
If f has a local maximum or minimum at c , then c is a critical number of f .



The extreme value theorem

1 Max & Min

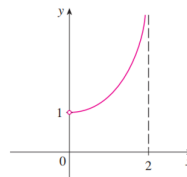
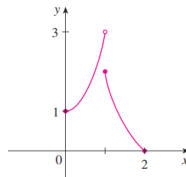
If f is **continuous** on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.



Example.

In the first figure, why isn't 3 the absolute maximum value?

In the second, does it have the absolute maximum and minimum value?



The closed interval method

1 Max & Min

To find the global maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of at the critical numbers of f in (a, b) .
2. Find the values of at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value, the smallest of these values is the absolute minimum value.

Example. Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1, \quad -0.5 \leq x \leq 4.$$

The closed interval method

1 Max & Min

To find the global maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

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Example. Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1, \quad -0.5 \leq x \leq 4.$$

Solution. Since f is continuous on $[-0.5, 4]$, we can use the CIM.

(1) Find critical numbers: $f'(x) = 3x(x - 2) = 0 \iff x = 0$ or $x = 2$

The values of f at these critical numbers are $f(0) = 1$, $f(2) = -3$

(2) The values of f at the endpoints of the interval are $f(-0.5) = 0.125$, $f(4) = 17$.

(3) Comparing these four numbers, we see that the absolute maximum value is $f(4) = 17$ and the absolute minimum value is $f(2) = -3$.



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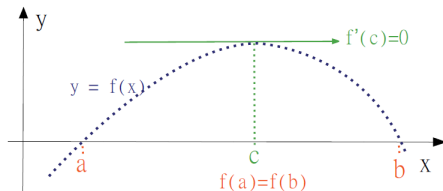
Rolle's theorem

2 The mean value theorem

Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$.

Then there is a number c in (a, b) such that $f'(c) = 0$.



Example. Find all number c that satisfy the conclusion of the Rolle's Theorem, $f(x) = \cos(2\pi x)$, on $[-1, 1]$.

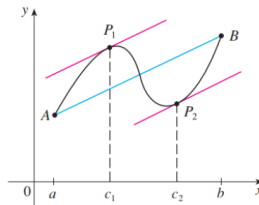
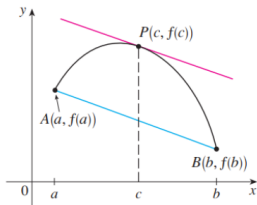
The mean value theorem

2 The mean value theorem

Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$



Example. Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all $0 \leq x \leq 2$. How large can $f(2)$ possibly be?

Solution

2 The mean value theorem

We are given that f is differentiable, therefore, f is continuous for any x .

Therefore, we can apply the Mean Value Theorem on the interval $[0, 2]$

There exists a number c such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

So, $f(2) = -3 + 2f'(c)$. Since $f'(x) \leq 5$ for any x , thus $f'(c) \leq 5$. Now, $f(2) = -3 + 2f'(c) \leq -3 + 10 = 7$. Thus, the greatest possible value for $f(2)$ is 7.



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3 Inflection Point

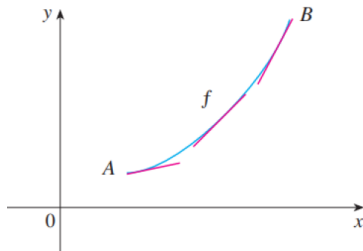
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Concave upward & Concave downward

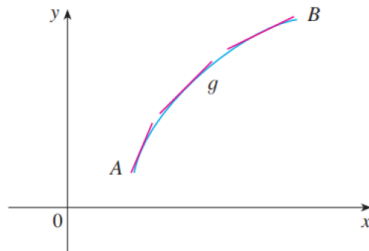
3 Inflection Point

Definition

If the graph of f lies above all of its tangents on an interval I , then it is called **concave upward** on I .
 If the graph of f lies below all of its tangents on I , it is called **concave downward** on I .



(a) Concave upward



(b) Concave downward

Inflection point

3 Inflection Point

Definition

A point $P(a; f(a))$ on a curve $y = f(x)$ is called an **inflection point** if f is continuous at a and the curve **changes** from concave upward to concave downward or from downward to concave upward at P .

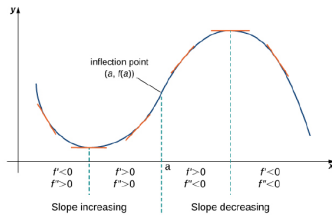
Find inflection points

Method 1:

Find x_0 such that $f''(x)$ does not exist, then x_0 is the inflection points.

Method 2:

1. Find x_0 such that $f''(x) = 0$.
2. At every x near x_0 that is found in Step 1, if $f''(x)$ changes the sign, then x_0 is an inflection point.



Example. Find inflection points if

$$f(x) = 3 + \sin x, [0, 2]$$



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Optimization Problems

4 Optimization Problems

Application in many areas of life:

- A businessperson wants to minimize costs and maximize profits.
- The traveler wants to minimize the transportation time.
- Problems such as maximizing areas, volumes, and profits and minimizing distances, times, and costs.

Example. We need to enclose a rectangular field with a fence. We have 500m of fencing material and a building is on one side of the field and will therefore not need any fencing. Determine the dimensions of the field that will enclose the largest area.

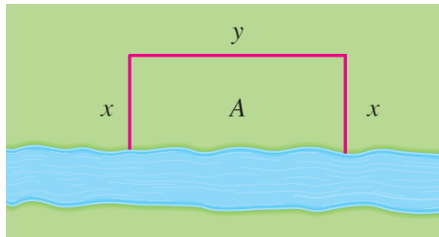




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Newton's method

5 Newton's Method

Newton's method is an iterative procedure used to find the roots of a function. Suppose we need to solve the equation $f(x) = 0$ and $x = r$ is the actual root of $f(x)$. We assume that the function $f(x)$ is differentiable in an open interval that contains r .

To find an approximate value for r :

1. Start with an initial approximation x_1 close to r .

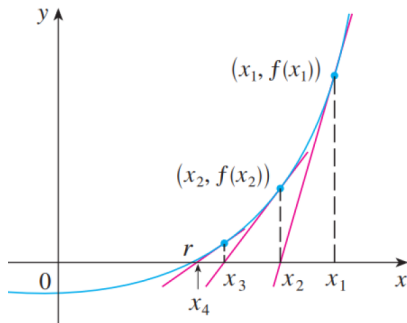
2. Determine the next approximation

by the formula

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}.$$

3. Continue the iterative process using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$





Newton's method - Example

5 Newton's Method

Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

Newton's method - Example

5 Newton's Method

Starting with $x_1 = 2$, find the third approximation x_3 to the root of the equation $x^3 - 2x - 5 = 0$.

Solution. Apply Newton's method with

$$f(x) = x^3 - 2x - 5 \text{ and } f'(x) = 3x^2 - 2$$

$$\Rightarrow x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

$$\text{With } n = 1, \text{ we have } x_2 = x_1 - \frac{x_1^3 - 2x_1 - 5}{3x_1^2 - 2} = 2 - \frac{2^3 - 2 \cdot 2 - 5}{3 \cdot 2^2 - 2} = 2.1$$

$$\text{With } n = 2, \text{ we obtain } x_3 = x_2 - \frac{x_2^3 - 2x_2 - 5}{3x_2^2 - 2} = 2.1 - \frac{2.1^3 - 2 \cdot 2.1 - 5}{3 \cdot 2^2 \cdot 1 - 2} \approx 2.0946.$$



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Antiderivative (Nguyên hàm)

6 Antiderivative

Definition

A function F is called an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x in I .

Theorem

If F is an antiderivative of f on an interval I , the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Example 1. Find f if $f'(x) = x\sqrt{x}$ and $f(1) = 2$.

Example 2. A particle moves in a straight line and has acceleration (gia tốc) given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.



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Problems

7 Problems

1. Find the absolute maximum and absolute minimum values of the function on the given interval
 - a. $f(x) = 3x^2 - 12x + 5$, $[0, 3]$
 - b. $f(x) = x\sqrt{4 - x^2}$, $[-1, 2]$
2. Find the critical numbers of the function
 - a. $f(x) = 5x^2 + 4x$
 - b. $f(x) = \frac{x - 1}{x^2 - x + 1}$
3. Find all numbers that satisfy the conclusion of the Rolle's Theorem.
 - a. $f(x) = x\sqrt{x + 2}$, $[-2, 0]$
 - b. $f(x) = (x - 2)x^2$, $[0, 2]$
4. Find all numbers that satisfy the conclusion of the Mean Value Theorem
 - a. $f(x) = 3x^2 + 2x + 5$, $[-1, 1]$
 - b. $f(x) = e^{-2x}$, $[0, 3]$
5. If $f(1) = 10$ and $f'(x) \geq 2, \forall x \in [1, 4]$, how small can $f(4)$ possibly be?
6. Find the inflection points for the function
 - a. $f(x) = x^4 - 4x + 1$
 - b. $f(x) = xe^x$



Problems

7 Problems

7. Find $f(x)$ for $f'(x) = \sqrt{2x+1}$ and $f(0) = 1$.
8. Use Newton's method with the specified initial approximation x_1 to find x_3 .
 - a. $f(x) = 6x^2 - 2x + 3$
 - b. $f(x) = \frac{x^2 + x + 2}{x}$.
9. Find the most general anti-derivative of the function.
 - a. $x^3 + 2x - 4 = 0$, $x_1 = 1$
 - b. $\ln(x^2 + 1) - 2x - 1 = 0$, $x_1 = 1$.
10. Find the anti-derivative of that satisfies the given function.
 - a. $f(x) = 5x^4 - 2x^5$, $F(0) = 4$
 - b. $f(x) = 4 - \frac{2x}{x^2+1}$, $F(0) = 1$.
11. A particle is moving with the given data. Find the position of the particle.
 - a. $v(t) = \sin t - \cos t$, $s(0) = 0$
 - b. $v(t) = 10 \sin t + 3 \cos t$, $s(\pi) = 0$.



Q&A

Thank you for listening!